Proton Magnetometer Review

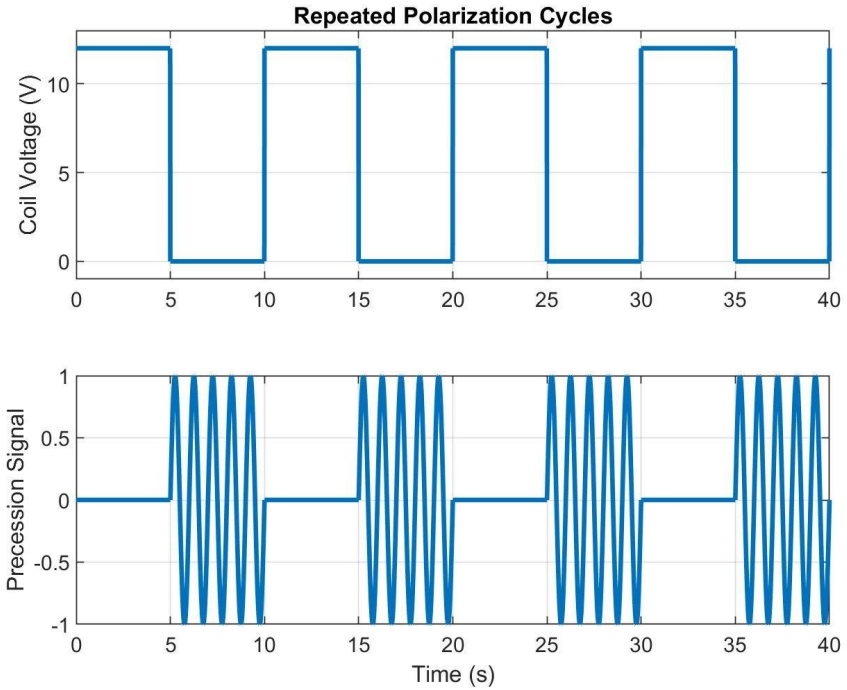
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Problem statement

- Find Vietnam-era aircraft wreckage, 2,500 to 4,000 kg of ferromagnetic material per fighter, in shallow coastal water.
- Sediment buries the target, so optics and sonar fail. A static magnetic field penetrates seawater without attenuation, so a surface-towed magnetic sensor still works at 0 to 100 m depth.
- Rule of thumb: 1,000 kg of steel gives about 1 nT of anomaly at 30 m.
- Detection ranges: 40 to 70 m for a whole aircraft at 0.25 nT, 25 to 45 m for major components at 4 nT.
- Commercial towed magnetometers cost $20K to $60K. Goal: a low-cost proton precession magnetometer towfish with better than 1 nT sensitivity.

What is a proton magnetometer

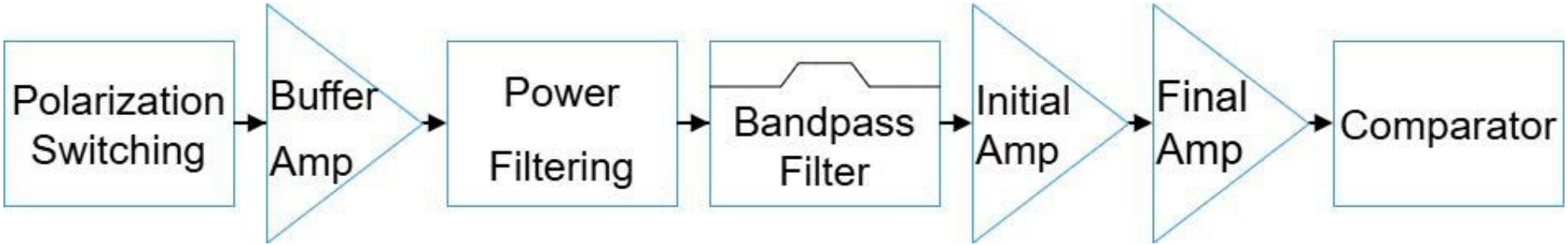
- A coil around a proton-rich fluid (water or kerosene) gets a DC pulse, which aligns the proton spins.
- The pulse switches off abruptly. The protons precess around Earth's field and induce a decaying sinusoid in a pickup coil, the free induction decay (FID).
- FID frequency is 1 to 2.8 kHz depending on local field strength.
- Field follows directly from frequency: B = f / γp, with γp about 42.577 Hz/nT.
- Sensitivity comes down to how precisely you can measure that frequency. 1 nT is about 0.04 Hz.



Previous converged design

All four SP26 teams ended up with the same architecture:

- Polarize: DC pulse through a coil around the fluid (24 V, 1.2 to 10 A, about 5 s cycle).
- Switch off fast, with flyback/TVS diodes and RC snubbers to clamp the back-EMF.
- Amplify and filter the FID: instrumentation amp front end, 0.85 to 5 kHz bandpass, blanking switch to protect the front end during the pulse.
- Measure frequency, then GPS-tag and log to SD card.
- Opposed or dummy coils cancel common-mode noise.



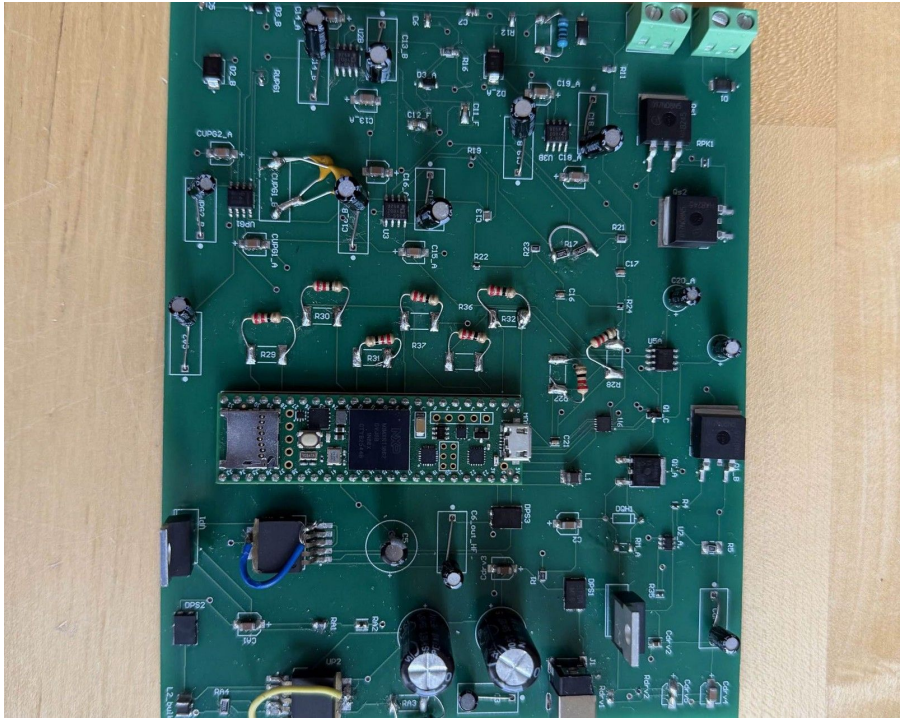
Completed work: polarization and coils

- Every team demonstrated MOSFET plus gate-driver pulsing. One hit a 4.7 A pulse with minimal ringing.
- Teams built 3D-printed coil forms, winding chucks, and turn counting jigs.
- Coil geometry iterated from toroid to solenoid to a final layout (3-coil, series-opposed pair, or bottle-in-bottle).
- Fluid sealing in printed forms took several epoxy and threaded-cap iterations.



Completed work: analog front end and frequency measurement

- Analog chains verified stage by stage with function generators: a 2000x chain (Finding Titan) and 0.3 mV to 1.5 V, about 5000x (Tow-fish & Furious).
- Teams moved from discrete op-amps to AD620/INA828 instrumentation front ends with a DG419 blanking switch.
- MCU timer counting and FFT proved too coarse, so teams built FPGA+TCXO reciprocal counting (2208 Hz in, 2207.99 Hz out), RP2040 PIO comparator counting, and zero-crossing interpolation.
- All teams logged SD-card CSV with GPS metadata and tested modular firmware against synthetic signals.



Open problem: no real FID was ever captured

- Every reported result came from a function generator, a driven test coil, or a hand-moved magnet. None came from protons in the team's own fluid.
- Each team proved the analog chain, comparator, and firmware in isolation on clean synthetic inputs. Nobody closed the link from fluid to microvolt FID to frequency.
- This is the largest gap in the prior work and the first thing to fix this semester: put a real coil and fluid on the scope and capture a genuine decaying FID before any integration work.

Open problem: noise floor and prototyping platform

- Breadboard and perfboard parasitics masked μV-level signals.
- Power-supply ripple fell inside the 2.1 kHz Larmor band, so filtering could not remove it.
- 60 Hz and about 60 kHz environmental pickup, plus the long tow cable acting as an antenna.
- Discrete op-amp chains had too high a noise floor and poor CMRR. Gain distribution was also wrong, with too much gain before the bandpass.
- Plan: leave breadboards early, star ground, shielded twisted coil leads, physical separation between coil and electronics.

Open problem: pulse-to-receiver coupling

- The high-current polarization transient rings and couples into the pickup coil, saturating the front end.
- Blanking timing is a knife-edge. Too short and the transient gets through; too long and the short-lived FID is gone.
- High-current loops near analog circuitry caused oscillation. Separation, shielding, and series-opposed dummy coils only partially helped.
- Plan: timed analog switch for blanking, and damp ringing at the source with TVS diode, snubber, and matched dummy coil.

Open problem: coil geometry and PCB bring-up

- Every team abandoned the toroid. It is impractical to wind, and polarization wants few thick turns while sensing wants many thin turns. No geometry was settled from first principles.
- Tow-fish & Furious put ten modules on one PCB and never finished debugging it: the -5 V rail clamped at 3.2 V, analog-rail capacitors were reversed, and a Teensy pin measured 1 ohm to ground.
- Hook Line & Towfish fixed ordering errors by scraping solder mask and bodge-wiring.
- Plan: pick one windable, sealable geometry, match opposed coils by measured resistance and inductance, use modular boards, and test power regulation standalone before fabrication.

Open problem: frequency precision and integration

- MCU gate counting gives 1 to 20 Hz accuracy; under 1 nT needs under 0.04 Hz.
- FFT resolution is 1/T, and the FID lasts 1 to 3 s, so about 1 Hz. Reaching 0.04 Hz would need a 25 s record.
- The FPGA, PIO, and zero-cross workarounds are sound but were only validated on clean synthetic waveforms, never on a real noisy decaying FID.
- Polarization, blanking, sampling, and GPS synchronization were left unfinished, with FPGA and Teensy desync causing repeated "measurement timed out" errors.
- Plan: validate the estimator against a traceable reference on real FID records, order long-lead parts early, and build the housing in parallel with the electronics.